Panel Interview Hub

AI Product Engineer - Round 1 | Product, policy, and data design

1. Student interview journey

State	What changes it	Student experience
Not eligible	Mandatory skill level is incomplete.	See missing requirements; booking stays locked.
Eligible and unbooked	All mandatory skills are verified.	See available slots and a clear booking call to action.
Booked / scheduled	Student commits an open slot.	See date, evaluator, meeting details, cancellation and reschedule rules.
Interview completed	Evaluator concludes the session.	See evaluation-pending status, never a blank result.
Pass	Evaluator submits a pass.	See badge award, result and student-facing feedback.
Retake later / fail	Evaluator records an outcome.	See feedback, attempt history and cooldown/retry guidance.
Under review / nullified	An integrity case is opened/resolved.	See neutral review notice, then a confirmed outcome or reinstatement.

Design principle: eligibility, booking, interview, evaluation and badge validity are separate state dimensions. A later integrity review can revoke a badge without destroying the original interview record.

2. Edge cases and assumptions

Case	Handling / assumption
Concurrent booking	The booking transaction locks the slot. First commit wins; the second student safely reselects.
Eligibility disputed after booking	Journey moves to review and operations pauses the booking without deleting history.
Interviewer unavailable	Preserve the booking, flag administration for reassignment, and notify the student after replacement.
Retake later	Prototype uses a 7-day cooldown and a maximum of three attempts; both are configurable policy.
Late cancellation or reschedule	Student self-service closes two hours before start. Overrides need an admin reason and audit entry.
No show / delayed evaluation	Attempt is retained, evaluator/admin queues show the exception, and the student sees a clear pending status.
Malpractice after award	Open a review; only an authorized admin can nullify or reinstate; preserve evidence and audit history.
Privacy	Students see public feedback only. Private notes and case evidence are role-scoped by row-level access.

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AI Product Engineer - Round 1 | Data model and prototype evidence

3. Data model

Identity and roles: *users* maps Supabase Auth identities to STUDENT, INTERVIEWER, and ADMIN. *students* and *interviewers* carry role-specific profile and capacity fields.

Eligibility: *tracks -> badges -> badge_requirements*, plus *student_skill_progress*, feed *student_badge_journeys*. The journey stores eligibility, badge validity, and attempt count.

Scheduling: *interviewer_availability*, *interviewer_unavailability*, and *interview_slots* describe capacity. *interviews* is one historical attempt per reservation, with scheduled times and evaluator.

Assessment and integrity: *interview_evaluations* stores Pass / Fail / Retake Later plus rubric, public feedback, and private notes. *malpractice_cases*, *badge_events*, *notifications*, and append-only *audit_logs* preserve the institutional record.

Guard	Implementation
No double booking	Filtered unique index on active interview slots, plus SELECT FOR UPDATE inside the booking RPC.
No overlapping active attempt	Filtered unique index on active journeys and overlap validation before insert.
Role protection	Auth-to-profile mapping, row-level policies, security-definer RPCs, and scoped student/interviewer/admin reads.
Auditability	Booking, evaluation, status, and integrity actions create immutable operational records.

4. Working prototype

The sign-in page has an **Assignment demo** path with seeded scenarios. It demonstrates: student eligibility and slot booking; interviewer availability, daily schedule, and outcome recording; student status/result screens; an admin eligible/booked/unbooked pipeline with reminders; and malpractice adjudication. The live student path uses Supabase Auth, PostgreSQL slots, realtime updates, and an atomic booking RPC.

5. Three important student messages

Moment	Exact message
Eligible	You have completed the requirements for this badge. Choose an available interview slot to continue.
Booked	Your interview is confirmed. Your evaluator, joining details, and preparation guidance are available in My Interviews.
Result / review	Your interview result is available. If your badge is under review, we will show the official decision here when the review is complete.

Before and after: the student home and badge-result views switch among incomplete, eligible, booked, evaluation pending, awarded, under review, and nullified states. The demo scenario switch makes each state visible during the walkthrough.